

Recommendation System for Personalized Content Discovery

Dataset - Netflix Prize Dataset

Objective -

Develop a personalized movie recommendation system capable of:

- Learning user preferences
- Predicting unseen ratings
- Generating Top-K recommendations
- Improving content discovery

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Problem Statement

Streaming platforms contain thousands of movies, making content discovery difficult for Users. Recommendation systems solve this problem by analyzing user behavior and suggesting relevant content.

Dataset

Netflix Prize Dataset

- 100,480,507 Ratings
- 480,189 Users
- 17,770 Movies
- Rating Scale: 1–5

Dataset Challenges

- Extremely sparse interactions
- Large-scale user-item matrix
- Long-tail user activity

Why Recommendation Systems?

- Improve user engagement
- Improve user retention
- Enhance user experience

Exploratory Data Analysis (EDA)

Key Findings-

Rating Distribution

- Most ratings are between 3 and 5
- Rating 4 is the most common

User Activity

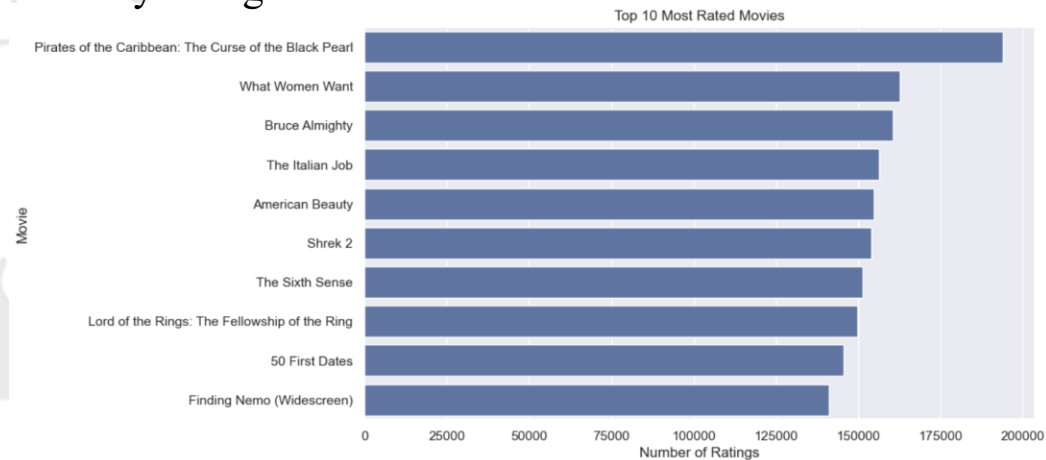
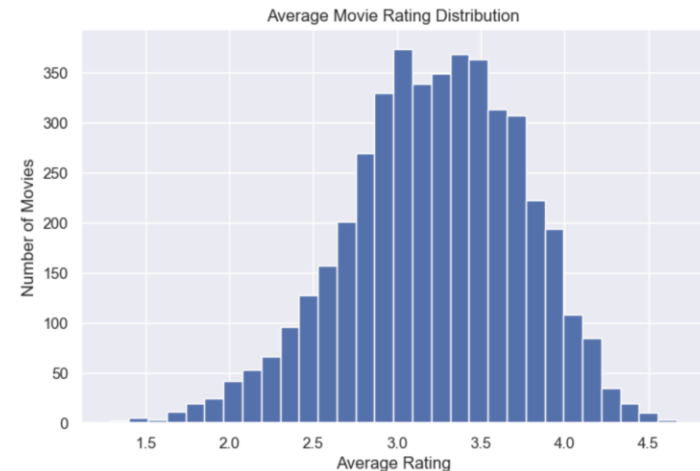
- Most users rate only a few movies
- Small number of highly active users contribute many ratings

Movie Popularity

- Few movies receive most ratings
- Popularity follows a long-tail distribution

Sparsity

- Dataset sparsity = 98.8%



Business Implication

Users have diverse preferences and limited interactions, making recommendation a challenging but valuable problem.

Methodology & Model Selection

Collaborative Filtering

- **Content-Based Filtering**- Uses movie features such as genre, actors, or directors.
- **Collaborative Filtering** -Uses user-rating behavior.

Since the Netflix dataset mainly contains: User IDs, Movie IDs, Ratings
Collaborative Filtering was the most suitable approach.

Model 1 : Item-Based Collaborative Filtering Principle

Recommends movies similar to those a user liked.

Implementation- KNN Basic, Cosine Similarity, Item-Item Similarity Matrix

Why Chosen- Simple baseline model, Easy to interpret, Widely used approach

Limitation- Needs sufficient rating overlap, Performance drops on sparse data

Model 2: Singular Value Decomposition (SVD) Principle

Learns hidden user and movie preferences through latent factors.

Why Chosen - Handles highly sparse data effectively, Captures hidden relationships between users and movies

Advantage - Better generalization, More accurate predictions on unseen ratings

Enhancing User Experience with Matrix Factorization

Matrix Factorization

Decomposes user-item interaction matrix into latent factors

Latent User Factors

Represents user preferences and behaviors

Latent Movie Factors

Captures movie characteristics and genres

Predicted Ratings

Estimates user ratings for unseen items

Recommendations

Suggests relevant items to users

Experimental Setup & Evaluation

Data Preparation

- Random sample extracted from dataset
- Users with fewer than 20 ratings removed
- Active Users: 1,926
- Final Dataset: 50,954 Ratings

Train-Test Split

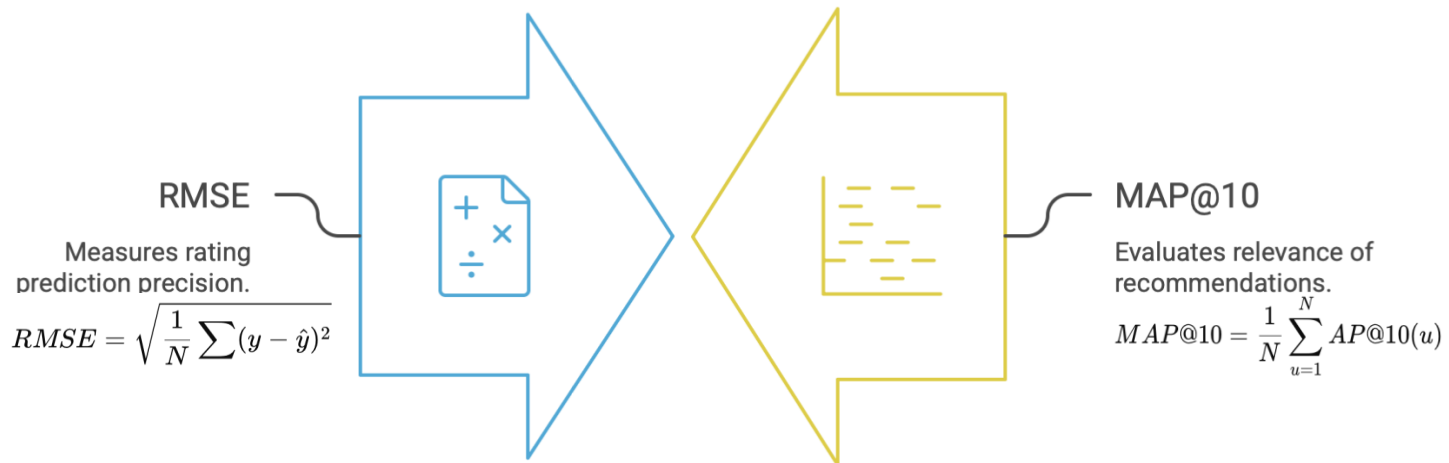
For each user:

- Historical ratings → Training Set
- Latest 5 ratings → Test Set

Training Ratings: 41,324

Test Ratings: 9,630

Evaluation Metrics



Why Both?

RMSE evaluates prediction quality. Lower the Better

MAP@10 evaluates recommendation quality. Higher the Better

Results & Model Comparison

MODEL PERFORMANCE

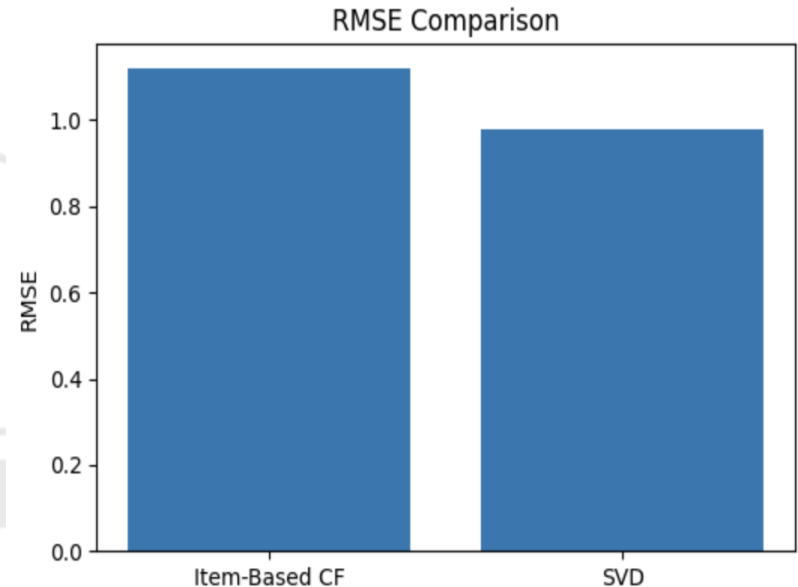
Model	RMSE
Item-Based CF	1.12
SVD	0.98

RANKING PERFORMANCE

Metric	Value
MAP@10	0.7615

Interpretation-

- **SVD achieved the lowest RMSE (0.98)**, indicating more accurate rating predictions than Item-Based CF (1.12).
- **MAP@10 = 0.7615** shows that relevant movies were generally ranked near the top of recommendation lists.
- SVD performed better because it learns **latent user preferences and movie characteristics**, making it more effective for sparse datasets.
- Therefore, **SVD was selected as the final recommendation model** for recommendation generation.



Recommendation Generation

Personalized Movie Recommendation Process

Watched Movies Identification

Recognizes movies the user has already seen

Unseen Movies Identification

Identifies movies the user hasn't watched yet

Rating Prediction

Estimates ratings for unseen movies using SVD

Movie Ranking

Orders unseen movies based on predicted ratings

Top-10 Recommendations

Presents the top 10 recommended movies to the user

Example User -

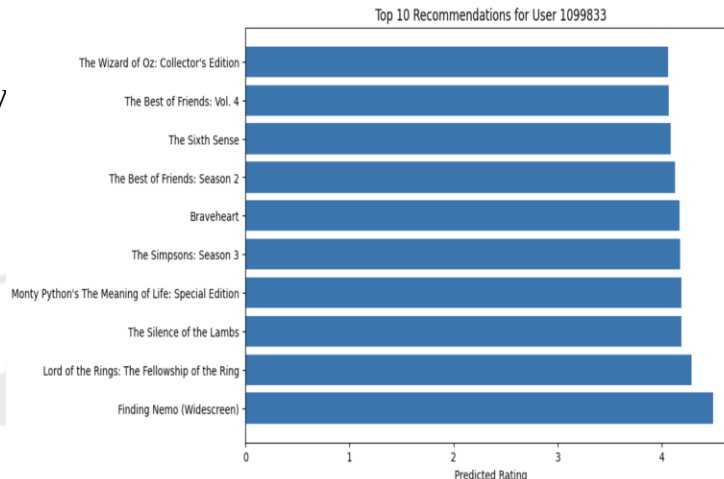
Highly rated movies:

- The Bourne Supremacy
- A Fistful of Dollars
- Urban Cowboy
- The Family Man

(Action, Drama, Thriller)

Recommended Movies:

- Finding Nemo
- Lord of the Rings
- The Silence of the Lambs
- Braveheart
- The Sixth Sense



Success Case:

Recommendations align with the user's preference for critically acclaimed and highly rated films.

Failure Case:

Some TV content such as:

- Friends
 - The Simpsons
- was recommended.

Reason:

Collaborative filtering relies solely on user interactions and does not explicitly use genre information.

Key Insights & Future Work

Key Insights-

Dataset sparsity was extremely high (**98.8%**)

User activity followed a **long-tail distribution**

Movie popularity was concentrated among a few titles

SVD outperformed Item-Based CF

- RMSE = **0.98**
- MAP@10 = **0.7615**

Personalized recommendations were successfully generated

Conclusion-

- Developed a personalized movie recommendation system using the Netflix Prize Dataset and compared Item-Based Collaborative Filtering with SVD.
- SVD achieved the best performance (**RMSE = 0.98, MAP@10 = 0.7615**) by effectively capturing latent user preferences.
- The system successfully generated meaningful Top-10 recommendations, demonstrating the effectiveness of recommendation systems for personalized content discovery.

Future Work-

- Hybrid Recommendation Systems
- Neural Collaborative Filtering
- Cold-Start Handling
- Explainable Recommendations
- Web Application Deployment